Quiz 1

Friday, September 4 — vectors, linear combinations, and Lab 0

Last updated: August 31, 2026.

Textbook: Boyd & Vandenberghe, *VMLS* (PDF)

Related lectures: Vectors, Arithmetic, Combinations, Inner product (stack, slice, dot product).

Related lab: Lab 0.

Schedule

Core

- B 1.1, 1.2, 1.6, 1.7, 1.9, 1.13, 1.18
- L Ch. 1 Ex. 2(a), linear-combination portion
- C0.1–C0.2: Enter, index, and modify a measurement vector (Octave or Python).
- C0.3–C0.4: Scale and add vectors; interpret in context.
- C0.5: Plot x, $2x$, and $-x$, and describe the geometric changes.
- C0.6: Predict an indexed output, then check (Boyd vs software indexing).

Stretch

B 1.5, 1.11, 1.20.

What a quiz item might change

The entries or units, a different linear combination, or four lines of Octave/Python whose resulting vector you predict.